

ZHENXING MI

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RESEARCH INTEREST

My research spans vision-language models, multimodal understanding and generation, video generation, and 3D/4D generation and reconstruction. I've published in top venues including CVPR, ICLR, ICML, TPAMI, TVCG, and AAAI, with contributions spanning multimodal understanding and generation, and Mixture-of-Experts (MoE) frameworks for scalable 3D scene modeling. Looking ahead, I'm excited to push the frontiers of vision-language models, multimodal understanding and generation models, video diffusion models, and 3D/4D world generation models, to build scalable generative systems that can perceive, reason, and create in dynamic environments.

EDUCATION

- **The Hong Kong University of Science and Technology** Feb. 2021 - June. 2026 (Expected)
Ph.D. candidate in Computer Science and Engineering
Advisor: Prof. Dan Xu
- **Huazhong University of Science and Technology** Sept. 2017 - June 2020
M.Sc. in Automation
Advisor: Prof. Wenbing Tao
- **Huazhong University of Science and Technology** Sept. 2013 - June 2017
B.Sc. in Control Science and Engineering
Advisor: Prof. Wenbing Tao

EXPERIENCE

- **Personalized Generative AI, Snap Research.** July. 2024 - Oct 2024
Research Intern, Multimodal generation
Mentor: Kuan-Chieh Wang, Guocheng Qian and Kfir Aberman
- **Applied Research Center (ARC), PCG, Tencent** Jan. 2024 - May 2024
Research Intern, 3D generation
Mentor: Xintao Wang

PUBLICATIONS

- **4D World Modeling via Video Generation:**
 - **Zhenxing Mi**, Yuxin Wang, Dan Xu. "One4D: Unified 4D Generation and Reconstruction via Decoupled LoRA Control.", arXiv:2511.18922, *Technical report*.
- **Multimodal Understanding and Generation:**
 - **Zhenxing Mi**, Kuan-Chieh Wang, Guocheng Qian, Hanrong Ye, Runtao Liu, Sergey Tulyakov, Kfir Aberman, Dan Xu. "I Think, Therefore I Diffuse: Enabling Multimodal In-Context Reasoning in Diffusion Models.", *ICML 2025*.
- **Scalable NeRF via Mixture of Experts (MoE):**
 - **Zhenxing Mi**, and Dan Xu. "Learning Heterogeneous Mixture of Scene Experts for Large-scale Neural Radiance Fields", *TPAMI 2025*.
 - **Zhenxing Mi**, and Dan Xu. "LeCO²-NeRF: Learning Continuous and Compact Large-Scale Occupancy for Urban Scenes.", arXiv:2411.11374, *Technical report*.
 - **Zhenxing Mi**, and Dan Xu. "Switch-NeRF: Learning Scene Decomposition with Mixture of Experts for Large-scale Neural Radiance Fields." *ICLR 2023*.
- **Efficient Multiview Stereo:**
 - **Zhenxing Mi**, Di Chang, and Dan Xu. "Generalized Binary Search Network for Highly-Efficient Multi-View Stereo." *CVPR 2022*.

■ Scalable Surface Reconstruction Network:

- Ganzhangqin Yuan*, Qiancheng Fu*, **Zhenxing Mi***, Yiming Luo*, and Wenbing Tao. “SSRNet: Scalable 3D Surface Reconstruction Network.” *IEEE TVCG* 2022.
- Yiming, Luo*, **Zhenxing Mi***, and Wenbing Tao. “DeepDT: Learning Geometry From Delaunay Triangulation for Surface Reconstruction.” *AAAI* 2021.
- **Zhenxing Mi***, Yiming Luo*, and Wenbing Tao. “SSRNet: Scalable 3D Surface Reconstruction Network.” *CVPR* 2020.

PATENTS

- **A local adaptive 3D point cloud denoising method based on depth maps.**

Inventor: Wenbing Tao, Zhenxing Mi

Authorization Publication Number: CN107845073A

Status: Authorized

Country/Region: China

AWARDS

- **2018:** Graduate School Scholarship (First Prize), Zhixing Scholarship (Third Prize), Outstanding Student
- **2017:** Outstanding Graduate, Graduate School Scholarship (First Prize)
- **2014:** National Encouragement Scholarship, by the Ministry of Education of China

SKILLS

- **Programming Languages:** Python, C++, CUDA
- **Deep Learning Framework:** Pytorch, Tensorflow
- **Operating Systems:** Linux (Ubuntu, CentOS), Windows, MacOS
- **Languages:** Chinese (native), English (fluent, academic speaking and writing)

SERVICE

- **Conference Reviewer:** CVPR, ECCV, ICCV, ICLR, ICML, NeurIPS, AAAI, ACM MM
- **Journal Reviewer:** TPAMI, IJCV, TVCG, TMM

TEACHING

- TA for Introduction to Computer Science - Fall 2022/23
- TA for Parallel Programming - Spring 2021/22
- TA for Discrete Mathematical Tools for Computer Science - Fall 2021/22